System (UC1)
searchPatient(searchCriteria) selectPatientNDService(patientID, medicalService) selectAppointment(professionalID, date, time) verifyReferral(referralID) confirmAppointment()

Contract CO1: searchPatient
Operation: searchPatient(searchCriteria: String)
Cross References: Use Case: Book Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.

Postconditions:
This operation doesn't change the state of the system.

Contract CO2: selectPatientNDService
Operation: selectPatientNDService(patientID: Integer, medicalService: String)
Cross References: Use Case: Book Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.
- *patientID* already exists in the database.
- *medicalService* already exists in the database.

Postconditions:
This operation doesn't change the state of the system.

Contract CO3: selectAppointment
Operation: selectAppointment(professionalID: Integer, date: Date, time: Time)
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *professionalID* already exists in the database.
- A time slot *ts* for that professional at the given time and date already exists in the database.
- *ts.availability* is set to available.

Postconditions:

- An appointment *a* was created. (instance creation)
- *a.status* was set to requested. (attribute modification)
- *a* was associated with the patient's *patientID*. (association formed)
- *a* was associated with the medical service's *medicalService*. (association formed)
- *a* was associated with the professional's *professionalID*. (association formed)
- *a* was associated with *ts*. (association formed)
- *ts.availability* was set to unavailable. (attribute modification)

Contract CO4: verifyReferral
Operation: verifyReferral(referralID: Integer)
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *referralID* already exists in the database.
- The medical service needs a referral.

Postconditions:

- *a* was associated with the referral's *referralID*. (association formed)

Contract CO5: confirmAppointment
Operation: confirmAppointment()
Cross References: Use Case: Book Appointment
Preconditions:

- An appointment *a* with status requested exists in the database.
- *a.status* is set to requested.
- *a* is associated with a patient that exists in the database.
- *a* is associated with a medical service that exists in the database.
- *a* is associated with a professional that exists in the database.
- *a* is associated with a time slot *t* that exists in the database.
- *a* is associated with a referral that exists in the database. (onl
- *t.availability* is set to available.

Postconditions:

- *a.status* was set to confirmed. (attribute modification)

System (UC2)
openPatientRecord(patientID) recordClinicalInformation(diagnosis, observation, treatment, followUpRequirements) issuePrescription(medications) requestFollowUp(followUpRequirements) endConsultation()

Contract CO6: openPatientRecord
Operation: openPatientRecord(patientID: Integer)
Cross References: Use Case: Patient Consultation
Preconditions:

- The user is authenticated, and their role allows access to clinical information.
- *patientID* already exists in the database.

Postconditions:

- A consultation *c* was created. (instance creation)
- *c.startTime* was set to the current time and date. (attribute modification)
- *c.status* was set to in progress. (attribute modification)

- *c* was associated with the patient's *patientID*. (association formed)
- *c* was associated with the professional's *professionalID*. (association formed)

Contract CO7: recordClinicalInformation
Operation: recordClinicalInformation(diagnosis: String, observation: String, treatment: String, followUpRequirements: String)
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- *c.diagnosis* was set to diagnosis via user input. (attribute modification)
- *c.observation* was set to observation via user input. (attribute modification)
- *c.treatment* was set to treatment via user input. (attribute modification)
- *c.followUpRequirements* was set to follow up requirements via user input. (attribute modification)

Contract CO8: issuePrescription
Operation: issuePrescription(medications: List[String])
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- A prescription *p* was created. (instance creation)
- *p* was associated with *c*. (association formed)
- *p.medications* was set to medications (attribute modifications)
- *p.issueDate* was set to the current time and date. (attribute modifications)

Contract CO9: requestFollowUp
Operation: requestFollowUp(followUpRequirements: String)
Cross References: Use Case: Patient Consultation
Preconditions:
- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:
- A follow up request *f* was created. (instance creation)
- *f* was associated with *c*. (association formed)
- *f.requirements* was set followUpRequirements. (attribute modifications)
- *f.status* was set to pending. (attribute modifications)

Contract C10: endConsultation
Operation: endConsultation()
Cross References: Use Case: Patient Consultation

Preconditions:

- Consultation *c* already exists in the database.
- *c.status* is set to in progress.

Postconditions:

- *c.endTime* was set to the current time and date. (attribute modifications)
- *c.status* was set to completed. (attribute modifications)

System (UC4)
searchPatient(searchCriteria) selectPatientAndService(patientID, medicalService) generateInvoice() recordPayment(invoiceID, amount, paymentMethod)

Contract C11: searchPatient
Operation: searchPatient(searchCriteria: String)
Cross References: Use Case: Payment Process
Preconditions:

- The user is authenticated, and their role allows appointment editing.

Postconditions:
This operation doesn't change the state of the system.

Contract C12: selectPatientAndService
Operation: selectPatientAndService(patientID: Integer, medicalService: String)
Cross References: Use Case: Payment Process
Preconditions:

- The user is authenticated, and their role allows appointment editing.
- *patientID* already exists in the database.
- *medicalService* already exists in the database.

Postconditions:
This operation doesn't change the state of the system.

Contract C13: generateInvoice
Operation: generateInvoice()
Cross References: Use Case: Payment Process
Preconditions:

- The patient is not insured so they must pay.

Postconditions:

- An invoice *i* was created. (instance creation)
- *i.total* was set to the service's fee. (attribute modifications)
- *i.status* was set to unpaid. (attribute modifications)
- *i.issueDate* was set to the current time and date. (attribute modifications)
- *i* was associated with the patient's *patientID*. (association formed)

Contract C14: recordPayment
Operation: recordPayment(invoiceID: Integer, amount: Double, paymentMethod: String)
Cross References: Use Case: Payment Process
Preconditions:

- Invoice *i* already exists in the database.
- *i.status* is set to unpaid.

Postconditions:

- A payment *p* was created. (instance creation)
- *p.amount* was set to amount. (attribute modifications)
- *p.paymentMethod* was set to paymentMethod. (attribute modifications)
- *p.date* was set to the current time and date. (attribute modifications)
- *p* was associated with *i*. (association formed)
- *i.status* was set to paid. (association formed)

System (UC3)
openSchedule() selectAppointment(appointmentID) rescheduleAppointment(appointmentID) selectNewTimeSlot(date, time) confirmReschedule() cancelAppointment(appointmentID)

Contract C15: openSchedule
Operation: openSchedule()
Cross References: Use Case:
Reschedule/Cancel Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.

Postconditions:
This operation doesn't change the state of the system.

Contract C16: selectAppointment
Operation: selectAppointment(appointmentID: Integer)
Cross References: Use Case Reschedule/Cancel Appointment
Preconditions:

- The user is authenticated, and their role allows appointment editing.
- *appointmentID* already exists in the database.
- Appointment has not already been cancelled

Postconditions:
This operation doesn't change the state of the system.

Contract C17: rescheduleAppointment
Operation: rescheduleAppointment(appointmentID)
Cross References: Use Case Reschedule/Cancel Appointment
Preconditions:

- An appointment *a* already exists in the database
- *a* has not been cancelled.
- *a* is associated with an existing timeslot *ts*

Postconditions:

- *a.status was updated to reflect the rescheduling. (attribute modification)*
- *ts.availability was set to available (attribute modification)*

Contract C18: selectNewTimeSlot
Operation: selectNewTimeSlot(date, time)
Cross References: Use Case: Reschedule/Cancel Appointment
Preconditions:

- An appointment has been rescheduled
- The new date and time has already been entered

Postconditions:

- This operation doesn't change the state of the system.

Contract C19: confirmReschedule
Operation: confirmReschedule()
Cross References: Use Case: Reschedule/Cancel Appointment
Preconditions:

- An appointment a is being rescheduled
- A new time slot *ts* exists in the schedule
- *ts.availability* is set to available
- Selected time slots respects the applicable scheduling constraints

Postconditions:

- An appointment *b* with was created.
- *b.date* is set to the selected date and time.
- *b.status* was set to confirmed
- *b* was associated with the patient's patientID
- *b* was associated with the medical service
- *b was associated with ts*
- *b* was associated with the existing referral (when applicable)
- *ts.availability* was set to unavailable.

Contract C20: cancelAppointment
Operation: cancelAppointment(appointmentID)
Cross References: Use Case: Reschedule/Cancel Appointment
Preconditions:

- The user is authenticated with their role allowing appointment editing
- Appointment *a* already existing in databases
- *a* IS NOT CANCELLED
- *a* is associated with an existing time slot *ts*.

Postconditions:

- *a.status* was set to cancelled
- *ts.availability* was set to available.

System (UC5)
searchPatient(searchCriteria) createReferral(patientID, specialtyName, reason, priority)

Contract C21: searchPatient
Operation: searchPatient(searchCriteria)
Cross References: Use Case: Request Referral
Preconditions:

- The user is authenticated and their role allows access to patient information.

Postconditions:
This operation doesn't change the state of the system.

Contract C22: createReferral
Operation: createReferral(patientID, specialtyName, reason, priority)
Cross References: Use Case: Request Referral
Preconditions:

- Healthcare professional is authenticated
- *patientID* exists
- *reason* exists (referral)
- *name* exists (Medical Speciality)

Postconditions:

- An referral *r* with was created.
- *r.reason* is set to the user output.
- *r.status* was initialized
- *r.priority* was set to the priority of user input.
- *r* was associated with the *patientID*
- *r was associated with* the healthcare specialist
- *r* was associated with the requested medical speciality
- *r was associated with* ReferralDB